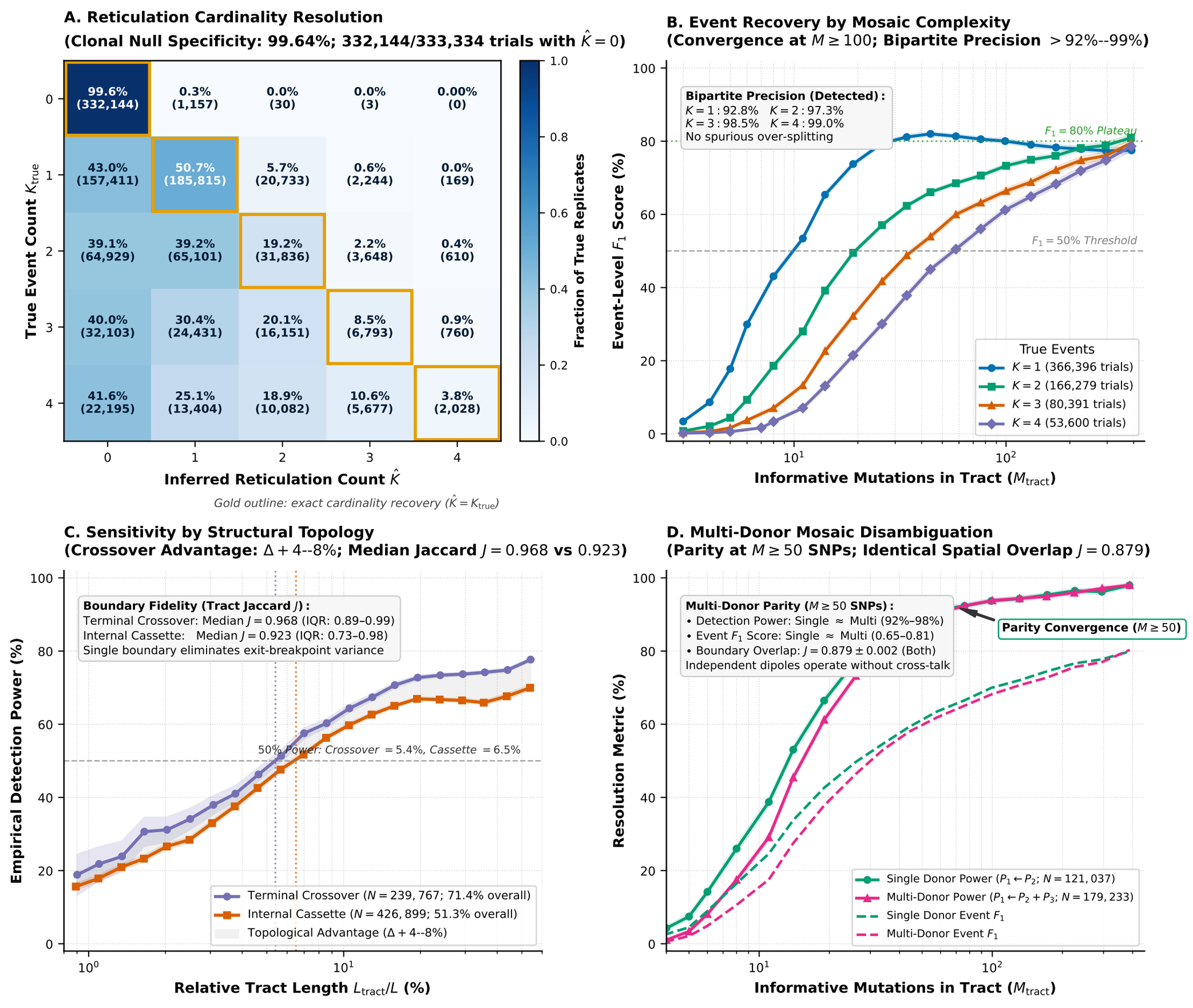

A. Reticulation Cardinality Resolution
(Clonal Null Specificity: 99.64%; 332,144/333,334 trials with $\hat{K} = 0$)
99.6% (332,144)
0.3% (1,157)
0.0% (30)
0.0% (3)
0.00% (0)
43.0% (157,411)
50.7% (185,815)
5.7% (20,733)
0.6% (2,244)
0.0% (169)
39.1% (64,929)
39.2% (65,101)
19.2% (31,836)
2.2% (3,648)
0.4% (610)
40.0% (32,103)
30.4% (24,431)
20.1% (16,151)
8.5% (6,793)
0.9% (760)
41.6% (22,195)
25.1% (13,404)
18.9% (10,082)
10.6% (5,677)
3.8% (2,028)
True Event Count K_{true}
Inferred Reticulation Count $\hat{K}$
Fraction of True Replicates
Gold outline: exact cardinality recovery ($\hat{K} = K_{true}$)
B. Event Recovery by Mosaic Complexity
(Convergence at $M \geq 100$; Bipartite Precision $> 92\%$--99%)
Bipartite Precision (Detected):
$K = 1$: 92.8% $K = 2$: 97.3%
$K = 3$: 98.5% $K = 4$: 99.0%
No spurious over-splitting
$F_1 = 80\%$ Plateau
$F_1 = 50\%$ Threshold
Event-Level F_1 Score (%)
Informative Mutations in Tract (M_{tract})
True Events
$K = 1$ (366,396 trials)
$K = 2$ (166,279 trials)
$K = 3$ (80,391 trials)
$K = 4$ (53,600 trials)
C. Sensitivity by Structural Topology
(Crossover Advantage: $\Delta + 4$--8%; Median Jaccard $J = 0.968$ vs 0.923)
Boundary Fidelity (Tract Jaccard J):
Terminal Crossover: Median $J = 0.968$ (IQR: 0.89–0.99)
Internal Cassette: Median $J = 0.923$ (IQR: 0.73–0.98)
Single boundary eliminates exit-breakpoint variance
50% Power: Crossover = 5.4%, Cassette = 6.5%
Empirical Detection Power (%)
Relative Tract Length L_{tract}/L (%)
Terminal Crossover ($N = 239,767$; 71.4% overall)
Internal Cassette ($N = 426,899$; 51.3% overall)
Topological Advantage ($\Delta + 4$--8%)
D. Multi-Donor Mosaic Disambiguation
(Parity at $M \geq 50$ SNPs; Identical Spatial Overlap $J = 0.879$)
Multi-Donor Parity ($M \geq 50$ SNPs):
• Detection Power: Single ≈ Multi (92%–98%)
• Event F_1 Score: Single ≈ Multi (0.65–0.81)
• Boundary Overlap: $J = 0.879 \pm 0.002$ (Both)
Independent dipoles operate without cross-talk
Parity Convergence ($M \geq 50$)
Resolution Metric (%)
Informative Mutations in Tract (M_{tract})
Single Donor Power ($P_1 \leftarrow P_2$; $N = 121,037$)
Multi-Donor Power ($P_1 \leftarrow P_2 + P_3$; $N = 179,233$)
Single Donor Event F_1
Multi-Donor Event F_1